

Nice work!

You've completed a reading.

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Applied Mathematical Methods for Computing

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Module 1

Algebra, vectors and matrices

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Module 2

Sequence and Series

▼

Module 3

Combinatorics

▼

Module 4

Probability and Statistics

▲

Lesson 4.1 Probability Definition and Properties

✓

Introduction to probability

Video • 11 min

✓

Probability

Practice Assignment • Grade: 100%

Lesson 4.2 Statistics

✓

Introducing statistics

Video • 10 min

✓

Worked examples (socks from a drawer)

Video • 7 min

✓

Sharing birthdays

Video • 6 min

✓

Statistics

Practice Assignment • Grade: 80%

Lesson 4.3 Summary and Assessments

✓

Analysing Real-World Data

Dialogue • 30 min

Looking back

Video • 2 min

✓

Check your understanding: End of module 4

Graded Assignment • Grade: 100%

Course Wrap-up

✓

Advanced Mathematical Methods for Computing: Course Summary

Reading • 10 min

The Ciphred Map: A Cartographic Escape Mystery

Role Play • 15 min

Course summary

Video • 56 sec

Advanced Mathematical Methods for Computing: Course Summary

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Congratulations on successfully completing the 'Advanced Mathematical Methods for Computing' course! We are incredibly proud of your dedication and hard work throughout this learning journey.

As we reach the end of our exploration of the advanced mathematical techniques essential for computer science, it's time to recap the key milestones, cement your understanding, and reflect on everything you've learned.

Algebra, Vectors and Matrices: You began by mastering the mathematical structures that underpin computational geometry and data transformations. You've learned to work with vectors and vector spaces, perform operations with matrices, and apply linear transformations. You've gained proficiency in using homogeneous coordinates for unified transformations and solving systems of linear equations with Gaussian elimination—skills essential for computer graphics, machine learning algorithms, and computational modeling.

Sequences and Series: You explored mathematical patterns through sequences and series, understanding convergence properties and summation techniques. You mastered the principle of mathematical induction for proving properties, and worked with recursive definitions and recurrence relations. These tools provide powerful methods for algorithm analysis, optimization, and understanding computational processes that build upon previous states.

Combinatorics: You developed a sophisticated toolkit for counting and arrangement problems by studying permutations, combinations, and fundamental counting principles. You applied the elegant Pigeonhole Principle to solve existence problems, and explored how combinatorial methods help analyze algorithm efficiency, data structures, and discrete computational systems.

Probability and Statistics: You concluded your journey by investigating uncertainty modeling and data analysis. You've learned to define sample spaces, compute probabilities including joint events, and calculate important statistical measures. These skills form the foundation for data science, machine learning, algorithmic analysis, and decision-making under uncertainty.

Throughout this course, you've built upon the foundations established in Mathematical Foundations for Computing, advancing your mathematical sophistication while maintaining a clear focus on computational applications. You've not only learned abstract mathematical concepts but also practiced applying them to solve real computational problems—connecting theory with practice in ways that enhance your capabilities as a computer scientist or computing professional.

This advanced mathematical toolkit prepares you for sophisticated challenges in algorithm design, artificial intelligence, data science, cryptography, and other cutting-edge areas of computer science within the Essential Mathematics for Computer Science specialisation.

If you enjoyed this course, we invite you to leave a review or provide feedback. Your insights are invaluable and will help us enhance the learning experience for future students.

Once again, congratulations on reaching this milestone, and thank you for being part of this learning journey. The advanced mathematical methods you've mastered will serve as powerful tools throughout your computing career.

Best of luck as you continue to explore the fascinating intersection of mathematics and computing!

Go to next item

✓ Completed